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Summary

Integrated M.S./Ph.D. researcher at the Korea University NLP Lab (Advisor: Prof. Heuseok Lim), specializing in large language models and natural language processing — with a focus on agentic LLMs, efficient reasoning, machine translation, information retrieval, and model merging. First-author research has been published at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**. Four papers were accepted at **EMNLP 2026** (2 Main, 2 Findings), with co-authored work at **ICLR 2026** and a series of preprints under review. Recipient of the **Best Paper Award (Top 1)** at HCLT 2025. Served as the post-training Project Manager and Korea University representative for **VAETKI**, Korea's national Sovereign AI Foundation Model. Author of 30+ peer-reviewed papers across top-tier NLP venues and journals.

Education

Integrated M.S./Ph.D. in Computer Science

Korea University, Seoul, Korea

2021 – Present

Expected: Feb. 2027

- NLP Lab, Advisor: Prof. Heuseok Lim.
- Research focus: large language models, natural language processing, agentic LLM, efficient reasoning, machine translation, information retrieval, and model merging.

Research Interests

Large Language Models · Natural Language Processing · Agentic LLM · Efficient Reasoning · Machine Translation · Information Retrieval · Model Merging

Work Experience & Major Projects

VAETKI — Korean Sovereign AI Foundation Model

Post-training Project Manager — Korea University Representative

Jun 2025 – Jan 2026

NC AI (NCSOFT)-led consortium

- Served as post-training PM representing Korea University in Korea's national **Sovereign AI Foundation Model** program, organized by the Ministry of Science and ICT — an approximately KRW 200 billion national initiative with staged six-month evaluations running through 2027.
- The Korea University team (Prof. Heuseok Lim) was selected as a core research partner in the NC AI (NCSOFT)-led grand consortium of 14 industry-academia-research institutions and 40 adopters (including POSCO DX, Lotte Innovate, KAIST, ETRI, and MBC).
- Led the full post-training pipeline — Supervised Fine-Tuning (SFT), Reinforcement Learning, and dataset curation — for the VAETKI model.
- **VAETKI model**: an industry-specialized sovereign foundation model, open-sourced on Hugging Face on Dec 30, 2025. A 100B+ parameter Mixture-of-Experts architecture (11B active parameters at inference) with a custom Multi-head Latent Attention (MLA) reducing memory usage by up to 83%.

- Achieved state-of-the-art results on Korean language and culture (CLICK), advanced reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [\[Model\]](#) [\[Technical Report\]](#)
- Press: [\[Korea University\]](#) [\[Yonhap News\]](#) [\[AI Times\]](#) [\[Newstomato\]](#)

Cross-lingual Language Transfer for Large Language Models

2024 – 2025

Lead Researcher (first author)

Published at ACL 2025 (Oral)

- Proposed a cross-lingual optimization method that efficiently transfers English-centric LLMs to new target languages while preserving source-language ability; accepted as an **Oral** presentation at ACL 2025.
- [\[PDF\]](#)

Efficient Reasoning for Large Reasoning Models

2025 – Present

Lead Researcher (first author)

Findings of EMNLP 2026; preprints under review

- Leading a first-author line of work on training-free adaptive thinking budgets (DART) and stable efficiency training via adaptive correct-only rewards (ACOER) for large reasoning models; DART was accepted to **Findings of EMNLP 2026**.
- Co-first author of **ThinkFuse**, a trajectory-aware test-time fusion method for small reasoning models, accepted to **Findings of EMNLP 2026**.
- [\[DART\]](#) [\[ACOER\]](#) [\[Code\]](#)

Awards

Best Paper Award (Top 1)

2025

Annual Conference on Human and Cognitive Language Technology (HCLT 2025)

For “*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*”; **J. Lee** (first author), M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim — Korea University; Soongsil University; Human-Inspired AI Research.

Publications

Underlined = author. * denotes equal contribution. Full profile on [Google Scholar](#).

2026

- “*The Geometry of Hallucination Detection: Why Simple Probes Are Sufficient*”
J. Lee, J. Seo, H. Lim — **EMNLP 2026** (Main)
- “*Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization*”
S. Lee, M. Kim, **J. Lee**, H. Lim — **EMNLP 2026** (Main)
- “*DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models*”
J. Lee, S. Hong, S. Lee, J. Seo, J. Son, S. Eo, C. Park, H. Park, H. Moon, H. Lim — **Findings of EMNLP 2026** [\[arXiv\]](#) [\[Code\]](#)
- “*ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models*”
M. Kang*, **J. Lee***, J. Seo, H. Lim — **Findings of EMNLP 2026** (*equal contribution)
- “*Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment*”
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim — **ICLR 2026** (Poster) [\[PDF\]](#)
- “*Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation*”
C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh — **Knowledge-Based Systems** [\[DOI\]](#)
- “*Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards*”
J. Lee, S. Lee, S. Hong, M. Kim, C. Park, H. Lim — Preprint, under review [\[arXiv\]](#) [\[Code\]](#)

- “*To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG*”
J. Lee, C. Park, H. Lim — Preprint, under review [\[arXiv\]](#) [\[Code\]](#)
- “*Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations*”
D.J. Lee*, J. Lee*, S. Lee, S. Hong, S. Son, S. Eo, J. Seo, H. Lim — Preprint, under review (*equal contribution) [\[arXiv\]](#) [\[Code\]](#)

2025

- “*Cross-lingual Optimization for Language Transfer in Large Language Models*”
J. Lee, S. Hong, H. Moon, H.S. Lim — **ACL 2025** (Oral) [\[PDF\]](#)
- “*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*”
J. Lee, M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim — Annual Conference on Human and Cognitive Language Technology (HCLT 2025) — **Best Paper Award**
- “*An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training*”
S. Son, C. Park, J. Lee, M. Shim, C. Lee, Y. Jang, J. Seo, J. Lim, H. Lim — **Expert Systems with Applications** [\[DOI\]](#)
- “*VAETKI Technical Report*”
NC AI Consortium — Technical Report [\[PDF\]](#)

2024

- “*Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation*”
J. Lee, H. Moon, S. Lee, C. Park, S. Eo, H. Ko, J. Seo, S. Lee, H.S. Lim — **Findings of ACL 2024** [\[PDF\]](#)
- “*Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0*”
C. Park, J. Lee, S. Son, K. Park, J. Jang, H. Lim — **Applied Sciences** [\[DOI\]](#)

2023

- “*PEEP-Talk: A Situational Dialogue-based Chatbot for English Education*”
S. Lee, Y. Jang, C. Park, J. Lee, J. Seo, H. Moon, S. Eo, S. Lee, B. Yahya, et al. — **ACL 2023** (System Demonstrations) [\[PDF\]](#)
- “*A Survey on Evaluation Metrics for Machine Translation*”
S. Lee, J. Lee, H. Moon, C. Park, J. Seo, S. Eo, S. Koo, H. Lim — **Mathematics** [\[PDF\]](#)
- “*Doubts on the Reliability of Parallel Corpus Filtering*”
H. Moon, C. Park, S. Koo, J. Lee, S. Lee, J. Seo, S. Eo, Y. Jang, H. Kim, H. Lee, et al. — **Expert Systems with Applications** [\[DOI\]](#)
- “*KoCheckGPT: Korean LLM Written Document Detector*”
M. Kang, J. Lee, S. Lee, S. Hong, J. Park, H. Lim — Annual Conference on Human and Language Technology (HCLT 2023) [\[PDF\]](#)

2022

- “*QUAK: A Synthetic Quality Estimation Dataset for Korean-English Neural Machine Translation*”
S. Eo, C. Park, H. Moon, J. Seo, G. Kim, J. Lee, H.S. Lim — **COLING 2022** [\[PDF\]](#)
- “*Empirical Analysis of Noising Scheme based Synthetic Data Generation for Automatic Post-editing*”
H. Moon, C. Park, S. Lee, J. Seo, J. Lee, S. Eo, H.S. Lim — **LREC 2022** [\[PDF\]](#)
- “*K-NCT: Korean Neural Grammatical Error Correction Gold-standard Test Set Using Novel Error Type Classification Criteria*”
S. Koo, C. Park, J. Seo, S. Lee, H. Moon, J. Lee, H. Lim — **IEEE Access** [\[PDF\]](#)
- “*Focus on FoCus: Is FoCus Focused on Context, Knowledge and Persona?*”
S.Y. Lee, J. Lee, C. Park, S. Eo, H. Moon, J. Seo, J. Park, H.S. Lim — 1st Workshop on Customized Chat Grounding Persona and Knowledge [\[PDF\]](#)
- “*There is No Rose Without a Thorn: Finding Weaknesses on BlenderBot 2.0*”
J. Lee, M. Shim, S. Son, C. Park, Y. Kim, H. Lim — arXiv preprint [\[arXiv\]](#)
- “*Language Chameleon: Transformation Analysis between Languages Using Cross-lingual Post-training*”
S. Son, C. Park, J. Lee, M. Shim, C. Lee, Y. Jang, J. Seo, H. Lim — arXiv preprint [\[arXiv\]](#)
- “*Korean and Multilingual Language Models Study for Cross-lingual Post-training (XPT)*”
S. Son, C. Park, J. Lee, M. Shim, C. Lee, K. Park, H. Lim — **Journal of the Korea Convergence Society** [\[PDF\]](#)

- “*Policy-based Performance Comparison Study of Real-time Simultaneous Translation*”
J. Lee, H. Moon, C. Park, J. Seo, S. Eo, S. Lee, S. Koo, H. Lim — **Journal of the Korea Convergence Society** [\[PDF\]](#)
- “*SaJuTeller: Conditional Generation Deep-Learning based Fortune Telling Model*”
H. Moon, **J. Lee**, J. Seo, S. Eo, C. Park, W. Kim, J. Park, H. Lim — Annual Conference on Human and Language Technology (HCLT 2022) [\[PDF\]](#)
- “*Korean Language Model Construction and Comparative Analysis with Cross-lingual Post-Training (XPT)*”
S. Son, C. Park, **J. Lee**, M. Shim, S. Lee, J.W. Lee, A. So, H. Lim — Annual Conference on Human and Language Technology (HCLT 2022) [\[PDF\]](#)
- “*DART: Data Augmentation using Retrieval Technique*”
S. Lee, J. Seo, **J. Lee**, M. Kang, H. Moon, C. Park, D. Jung, J. Lee, K. Park, *et al.* — Annual Conference on Human and Language Technology (HCLT 2022) [\[PDF\]](#)

2021

- “*Empirical Study on BlenderBot 2.0's Errors Analysis in Terms of Model, Data and Dialogue*”
J. Lee, S. Son, M. Shim, Y. Kim, C. Park, A. So, J. Park, H. Lim — **Journal of the Korea Convergence Society** [\[PDF\]](#)